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(1) Research Program/Project/Study Title: Acceptability of Broad Geloina (<i>Geloina Expansa</i>) and Greasyback Shrimp (<i>Metapenaeus ensis</i>) Pastil as a Rice Toppings	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Roxas
Province:		Capiz
Region:		VI
(5) Classification Research: Development:		(6) Mode of Implementation ☐ Single Authorship/Proponent

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☐ Basic ☒ Applied	☐ Pilot Testing ☐ Tech. Promotion/Commercialization	☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization
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(9) Rationale *(State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)*

Rice is a significant staple food source for more than two third of the world's population (Bhat and Riar, 2015), primarily consumed as a whole-grain cereal after cooking (Sasaki et al., 2015). Filipinos love eating rice (Sofista et al., 2020). Eating rice without viand or main dish can seem unsatisfying and bland. There are many different ways to make eating rice without viand more enjoyable and satisfying.

Seafood products, including crustacean shellfish, are recommended for human diet due to their health-promoting characteristics (Skonberg and Perkins 2002). In terms of the amount of fat and the proportions of saturated, monounsaturated, and polyunsaturated fat, shellfish provide a healthful diet for humans (Dong 2001).

Pastil or pastel, is a Filipino packed rice dish made with steamed rice wrapped in banana leaves with dry shredded beef, chicken, or fish. It originates from the Maguindanao and Cotabato people and is a popular cheap breakfast meal in Mindanao, especially among Muslim Filipinos (Punzalan 2017). Although traditional pastil is made from meat of beef and chicken, pastil it may also incorporate using seafood. Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) as a main ingredient to develop healthier and low cost pastil.

This study intends to promote the consumption Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) Pastil. Overall, the creation of pastil has the potential to provide customers with an innovative and appealing main dish that could also offer a new way to enjoy pastil.

(9-A) Project Description *(Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)*

The present study will utilize Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) as the main ingredient in making Patil. Other ingredients such as vegetable oil, onion, garlic, black pepper, salt, black pepper, sugar, vinegar, soy sauce and bay leaves will be used. The acceptability of the Pastil will be based on sensory qualities such as color, odor, taste, and texture. The best formulation will be used and undergo for consumer acceptability.

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Completely Randomized Design will be used with a single factorial design with four (4) treatments and three (3) replications. Analysis of Variance will be the statistical tool to use.

This will be conducted at the FFRDC laboratory for 5 months including the Preparation of materials, product formulation and evaluation, statistical analysis, consumer acceptability, and write-ups.

(10) Objectives/Statement of the Problem *(for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)*

This study aims to utilize Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) to produce Pastil as a Rice toppings. Specifically, it aims to:

1. To determine the acceptability of Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) pastil;
2. To evaluate the sensory properties of Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) pastil;
3. To assess consumer acceptability of the optimized product; and
4. To determine the Proximate Analysis of Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*).

(11) Review of Literature/Prior Art Search *(Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).*

Pastil, also spelled as pastel in Maguindanaon, patel or pater in Maranao and paster in Iranun, according to Polistico (2010). Pastil is a rice and viand rolled in a banana leaf which is similar to the 'binalot' of Tagalogs. Pastil is a Halal food made of boiled rice and in the middle topped with 'kagikit' – sautéed shredded meat of chicken, beef, or fish that is wrapped with a wilted banana leaf.

CN1965685A is an invention relates to a method for making spicy cabbage pork rice.

Present invention is realized by the following methods: (1) The rice is cooked in a rice cooker and used, (2) Weigh two or two pickled spicy cabbages for later use, (3) Take half a catty of pork and cut into pieces, add onion ginger garlic and seasoning to stew, then add spicy cabbage, (4) Add cooked pork and spicy cabbage to rice and stir.

CN1965687A Method for cooking rice with mushroom and chicken filets.

Disclosed is a process for preparing rice meal containing mushroom and chicken cutlet, which comprises the steps of cooking rice with electric cooker, slicing 100g of mushrooms, subjecting two chicken cutlets to stewing together with green onion, ginger and garlic till half-cooked, charging mushroom slices and stewing, mixing cooked mushrooms and chicken cutlet with rice and stirring. The beneficial effects of the invention include featured multiple constituents, rich and balanced nourishment, and unique taste.

This study relates to the two patented inventions based on the process and compositions of rice toppings. However, Broad Geloina (*Geloina Expansa*) and Greasyback shrimp (*Metapenaeus ensis*) pastil will be used as the main ingredients.

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)

INPUT	PROCESS	OUTPUT
Broad Geloina (<i>Geloina Expansa</i>) and Greasyback Shrimp (<i>Metapenaeus ensis</i>) and other ingredients	- Product Preparation and formulation of Pastil - Consumer Testing and sensory evaluation	- Optimized formulation of Broad Geloina (<i>Geloina Expansa</i>) and Greasyback Shrimp (<i>Metapenaeus ensis</i>) Pastil - Proximate Analysis of best formulation

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The study will be experimental research. It will focus on Acceptability of Broad Geloina (*Geloina Expansa*) and Greasyback Shrimp (*Metapenaeus ensis*) Pastil and conduct the exact proportion.

Table 1: Treatments of Broad geloina and Greasyback shrimp Pastil

Treatment A	Treatment B	Treatment C	Treatment D
100% Broad geloina	75% Broad geloina	50% Broad geloina	25% Broad geloina
	25% Greasyback shrimp	50% Greasyback shrimp	75% Greasyback shrimp

**Other ingredients are consistent.*

A. Materials and Ingredients

The materials to be use in this study are chopping board, knife, mixing bowl, mixing spoon, mixing cups and cooking equipment. The ingredients to be use are Broad geloina , Greasyback shrimp, vegetable oil, onion, garlic, black pepper, salt, black pepper, sugar, vinegar, soy sauce and bay leaves.

B. Methods

Preparation of Broad geloina

Depurate the Broad geloina for 24 hours, wash thoroughly in running water, and steam for 30 min. Remove the Broad geloina from the shell, clean the intestine and chop. Then, set aside.

Preparation of Greasyback shrimp

Wash the Greasyback shrimp. Add 1 cup of water to a pan and add the Greasyback shrimp. Lower the fire, boil for 12 mins per 1kg, then leave to cool. Remove the meat from the shell and chop. Then, set aside.

Cooking Procedures

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1. preheat the pan and add vegetable oil over low heat. Stir the garlic and onion.

1. Add Broad geloina and Greasyback shrimp and stir.
2. Season with salt, sugar, and pepper. Add bay leaves and soy sauce. Cook for 10 mins.
3. Then add vinegar and cook for another 10 mins.
4. Cool completely before storing in a bottle.

Experimental Design

This study will make use the experimental method or procedure involving the control or manipulation of conditions for the purpose of study the relative effects of various treatments applied to a number of samples, or of the same treatment applied to a number of different samples. A completely randomized block design (CRD) will be use during the conduct the sensory evaluation.

Sensory evaluation

The Broad geloina and Greasyback shrimp pastil will be prepared in accordance with the protocol for sensory evaluation.

Scorecards that contain the 9-point Hedonic Scale for acceptability will be used to evaluate the development of the product, 15 evaluators will evaluate the product. Only sensory characteristics such as color, taste, flavor, texture, and general acceptability will be evaluated.

For the consumer's preference, 100 respondents of varying ages and sex will evaluate the product.

Proximate analysis

Proximate composition of developed plant-based meat analogue carried out by standard AOAC method (AOAC 1995).

The Broad geloina and Greasyback shrimp pastil will be brought to the DOST laboratory for proximate composition analysis.

Data Collection

The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Scoring Variables

The Criteria for evaluation as to sensory characteristics of the product using a 9-point hedonic scale in terms of color, taste, flavor, texture, and general acceptability. The rating for general acceptability on 9-point hedonic scale is 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item was interpreted based on the result of the computation as to color, taste, flavor, texture, and general acceptability using the following scoring interval and verbal interpretation.

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Statistical analysis

The data collected on sensory evaluation was statically analyzed by using SPSS software. The analysis of variance (ANOVA) was used to analyze sensory evaluation results with the significant difference among the sample with different treatments.

(14) Major Activities/Work plan (*Enumerate in chronological order the task to be undertaken. Use gantt chart.*)

Research Objectives	Duration (months)				
	1	2	3	4	5
1. Preparation and selection of Raw Materials					
2. Product Development					
3. Product Evaluation					
4. Shelf life Testing					
5. Statistical Analyses					
6. Manuscript preparation and write-up					

(15) Expected Output (*Indicate the specific products, processes or services which the project is expected to produce; quantify when possible*)

1 Optimized Broad geloina and Greasyback shrimp pastil
1 patented product
1 IEC material

(16) Technology Transfer Activity Design (*Indicate the specific products, activity design, processes or services to be transferred as extension activities*)

Training on the process of Broad geloina and Greasyback shrimp Pastil

(17) Target Beneficiaries (*Who the clientele are and what are the expected outcome/effects of the use of the projects output*)

- **Broad geloina and Greasyback shrimp Farmers**
- **Food processors**
- **Students**
- **Entrepreneurs**
- **Public Consumer**

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(18) Personnel Requirement		
Name	Role (Program, Project, Study Leader)	Field of Specialization
STEPHANIE S. PIMENTEL	Study Leader	Biology
VIALYN B. REGALADO	Co-researcher	Food Science
Percent time devoted to the Program/ Project/ Study		
Program/Project Leader (in hours):		
Program/Project Members (in hours):		
(19) Budgetary Requirement		
Ingredients	Quantity	Amount
Broad geloina	30 kgs	3,000.00
Greasyback shrimp	30 kgs	4,500.00
Other Ingredients (vegetable oil, onion, garlic, black pepper, salt, black pepper, sugar, vinegar, soy sauce and bay leaves)		1,500.00
Gas during product Development.		1,300.00
Plastic cups for Sensory	5 packs	700.00
Proximate Analysis (moisture, ash, Protein, and Carbohydrates)		10,000.00
Transportation Allowance		1,200.00
Patent		5,000.00
TOTAL		27,200.00
(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)		
Abdullah, A. B., Ito, S., & Adhana, K. (2006, March). Estimate of rice consumption in Asian countries and the world towards 2050. In Proceedings for Workshop and Conference on Rice in the World at Stake (Vol. 2, pp. 28-43). Punzalan, Noel. 2017. "Pastil: Maguindanaon food on Christmas". Philippine News Agency. Republic of the Philippines. http://www.pna.gov.ph/articles/1019321 Accessed: October 22, 2018 Kuley, E., Özoğul, >Fatih, Özogul, Y., & Olgunoglu, A. I. (2008). Comparison of fatty acid and proximate compositions of the body and claw of male and female blue crabs (<i>Callinectes sapidus</i>) from different regions of the Mediterranean coast. International Journal of Food Sciences and Nutrition, 59(7-8), 573–580. doi:10.1080/09637480701451201 Bahtiar, Muhammad Fajar Purnama, Maruf Kasim & Ermayanti Ishak (2023) Reproductive biology of mangrove clams <i>Geloina expansa</i> (Mousson, 1849) from mangrove at Kendari Bay, Southeast Sulawesi, Indonesia, Marine Biology Research, 19:1, 48-58, DOI: 10.1080/17451000.2023.2185639		
(21) Capsule Curriculum Vitae (One-page brief C.V. only)		